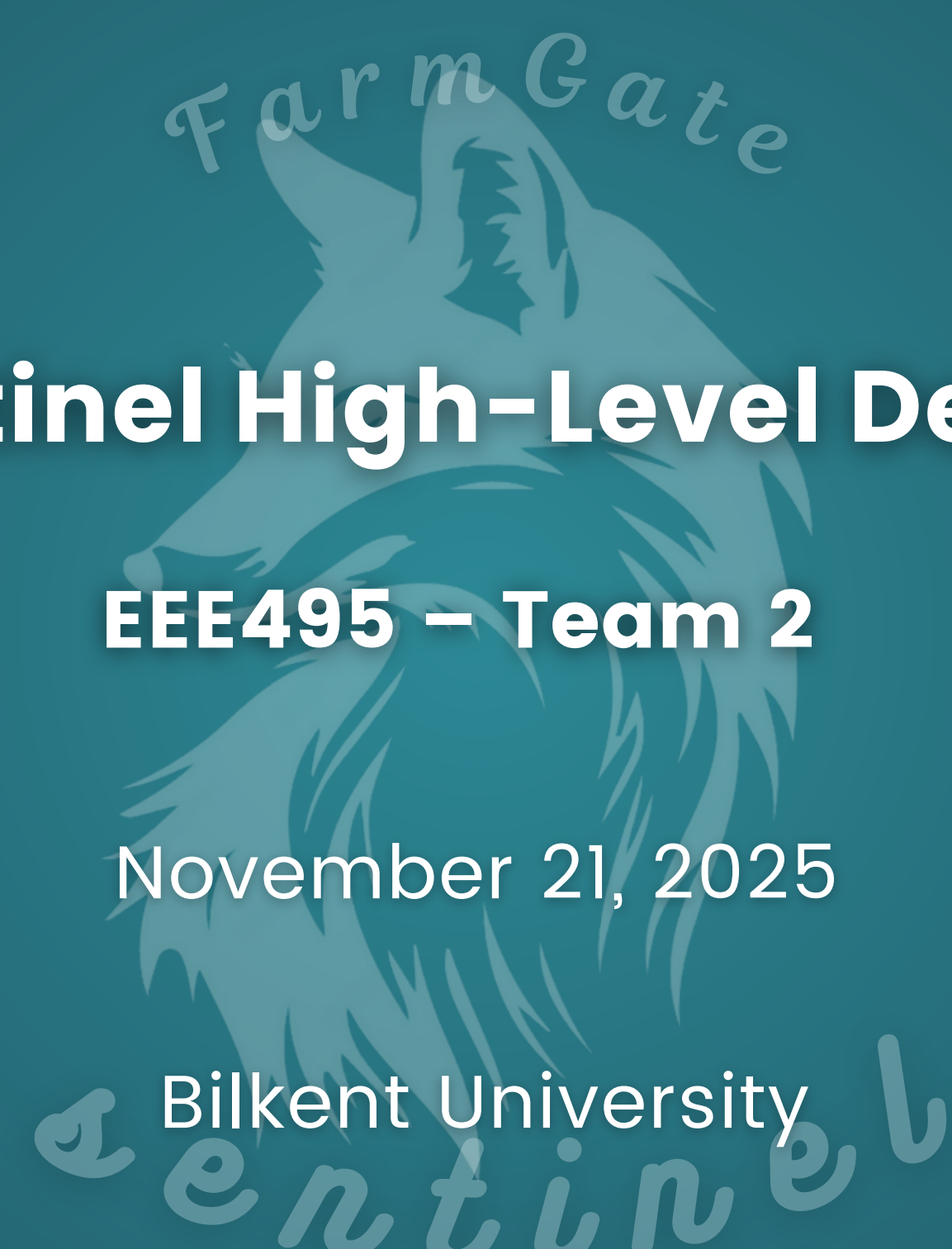




FarmGate Sentinel High-Level Design Overview

EEE495 – Team 2

November 21, 2025

Bilkent University

Functions and Brief Description of Product

- Smart wildlife-detection and deterrence system for farms
- Uses camera-based edge nodes and a local Raspberry Pi gateway
- Performs AI-based animal detection using YOLOv11
- Sends real-time alerts to the cloud and mobile app
- Provides remote deterrent actions and live dashboard with logs, images, and live info
- Designed for reliable operation even with weak connectivity

System Architecture

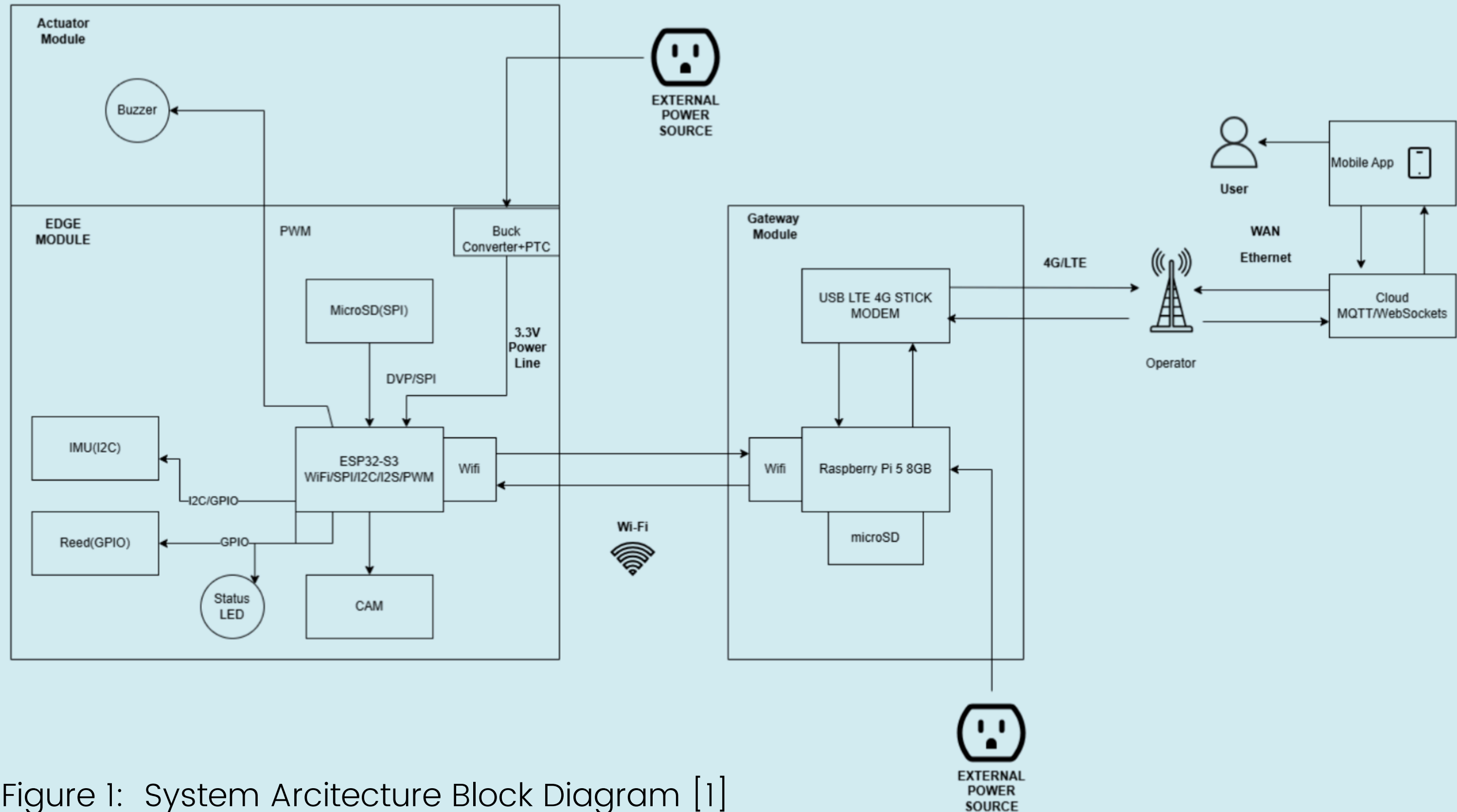


Figure 1: System Architecture Block Diagram [1]

Edge Node Hardware

Individual Blocks

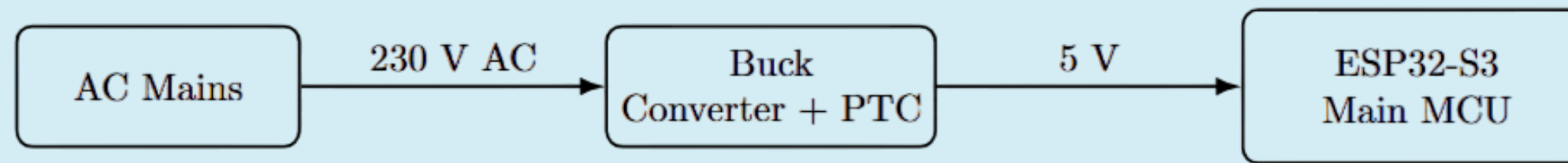


Figure 2: Power-supply subassembly

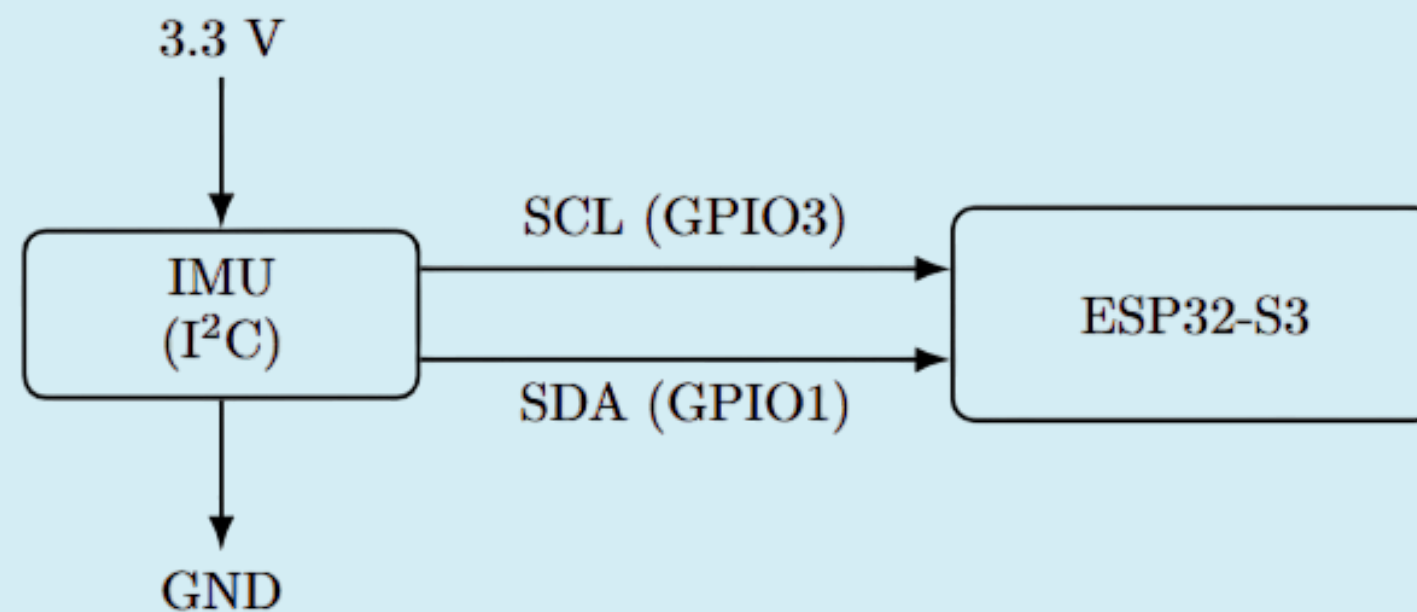


Figure 3: MU-ESP32 subassembly (I2C)

Edge Node Hardware Individual Blocks

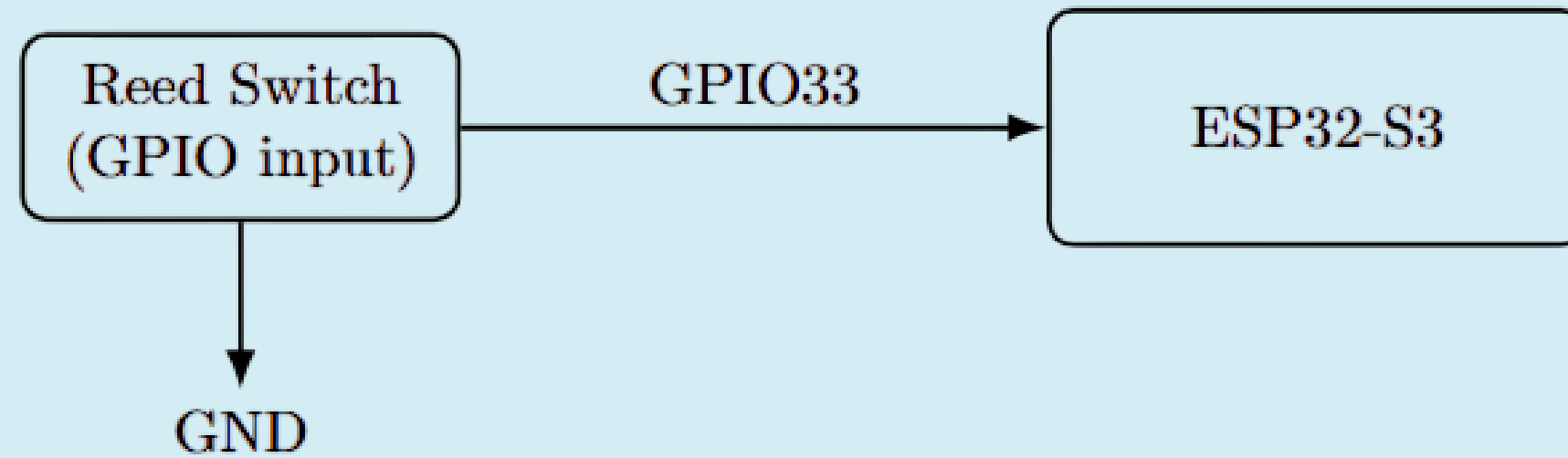


Figure 4: Reed-switch subassembly

(Series resistor not shown)

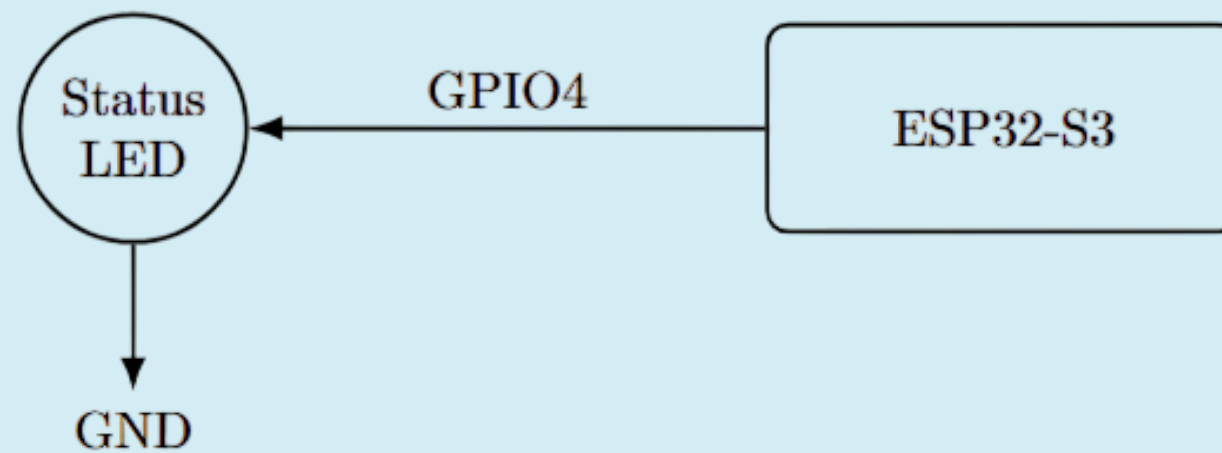


Figure 5: Status LED subassembly

Edge Node Hardware

Individual Blocks

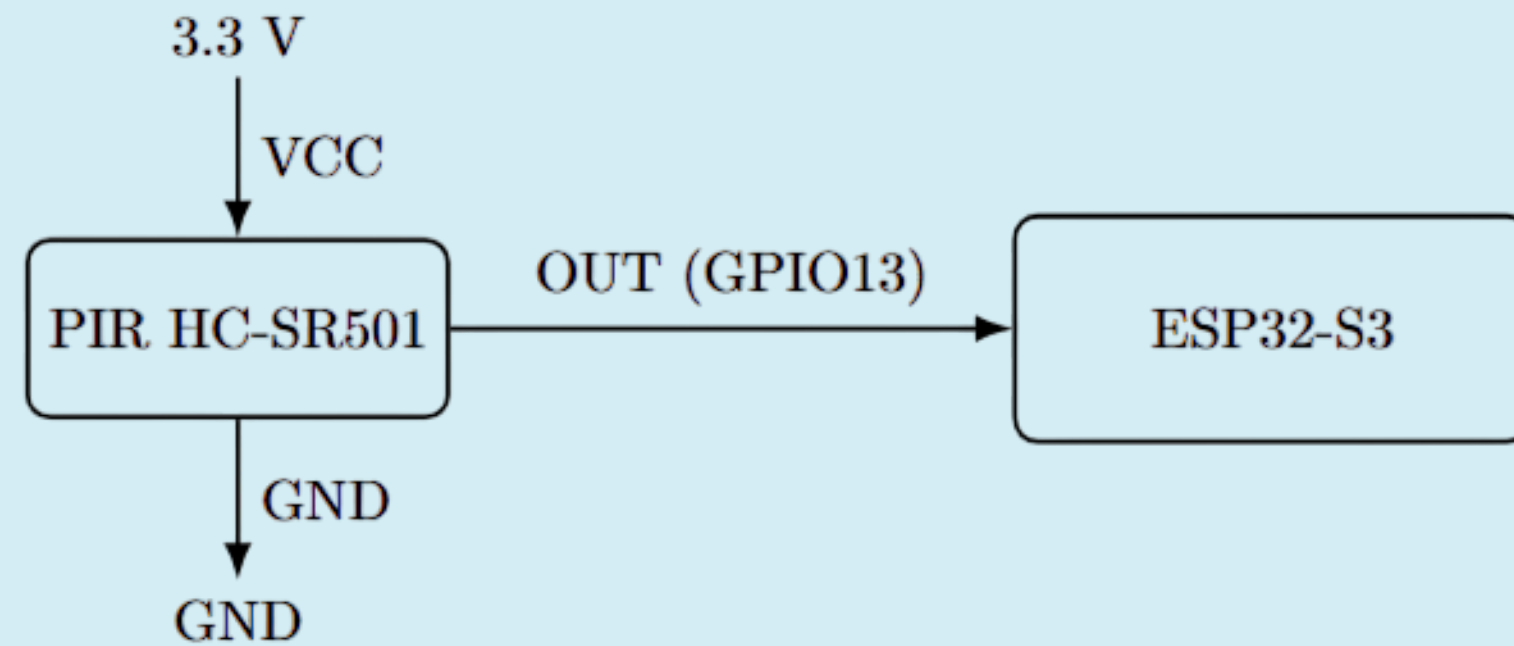


Figure 6: PIR motion-sensor subassembly

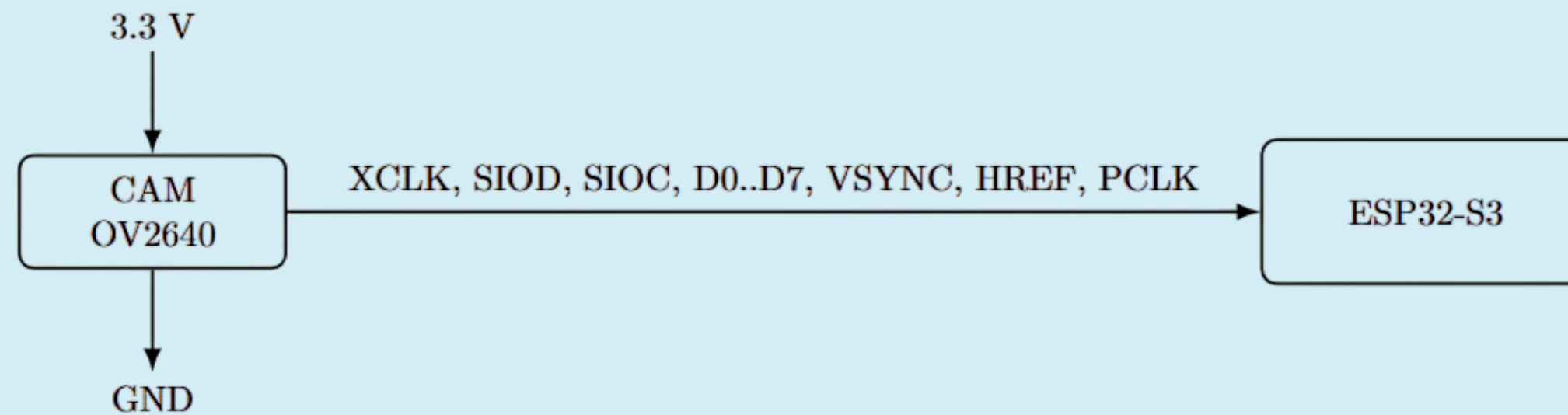


Figure 7: Camera subassembly

Edge Node Hardware

Individual Blocks

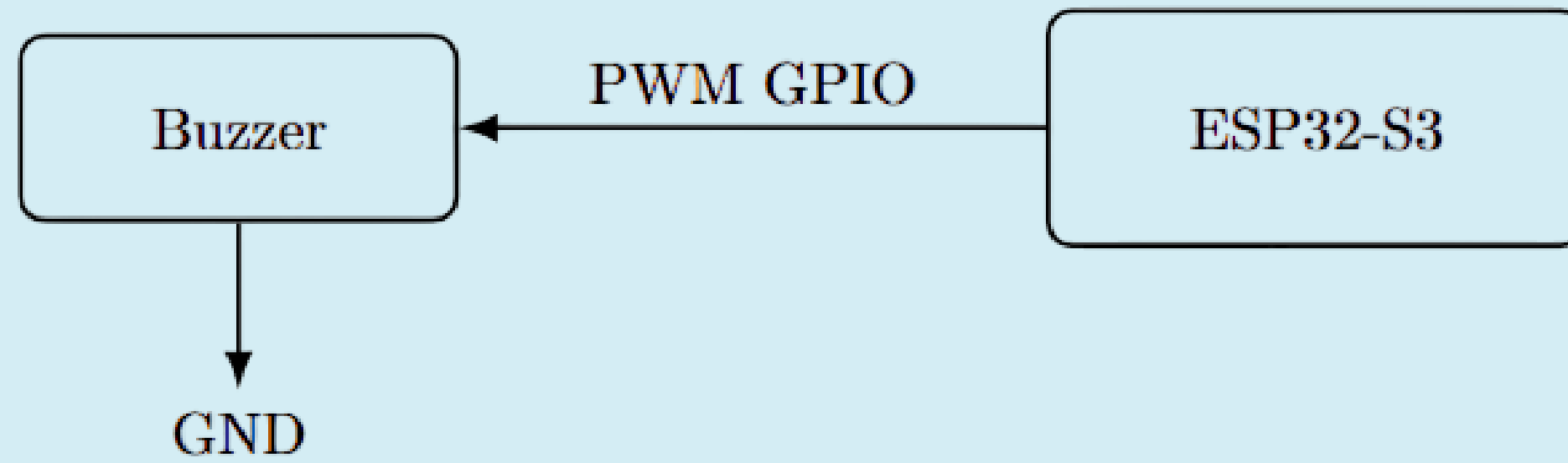


Figure 8: Buzzer subassembly

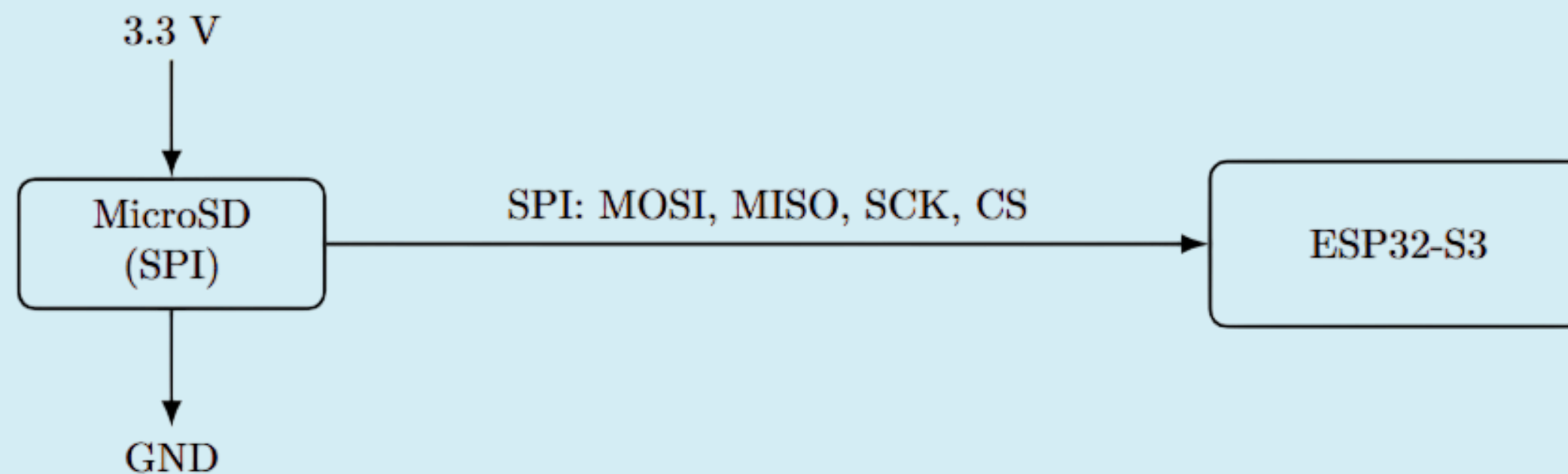


Figure 9: MicroSD subassembly

Edge Node Hardware Block Diagram

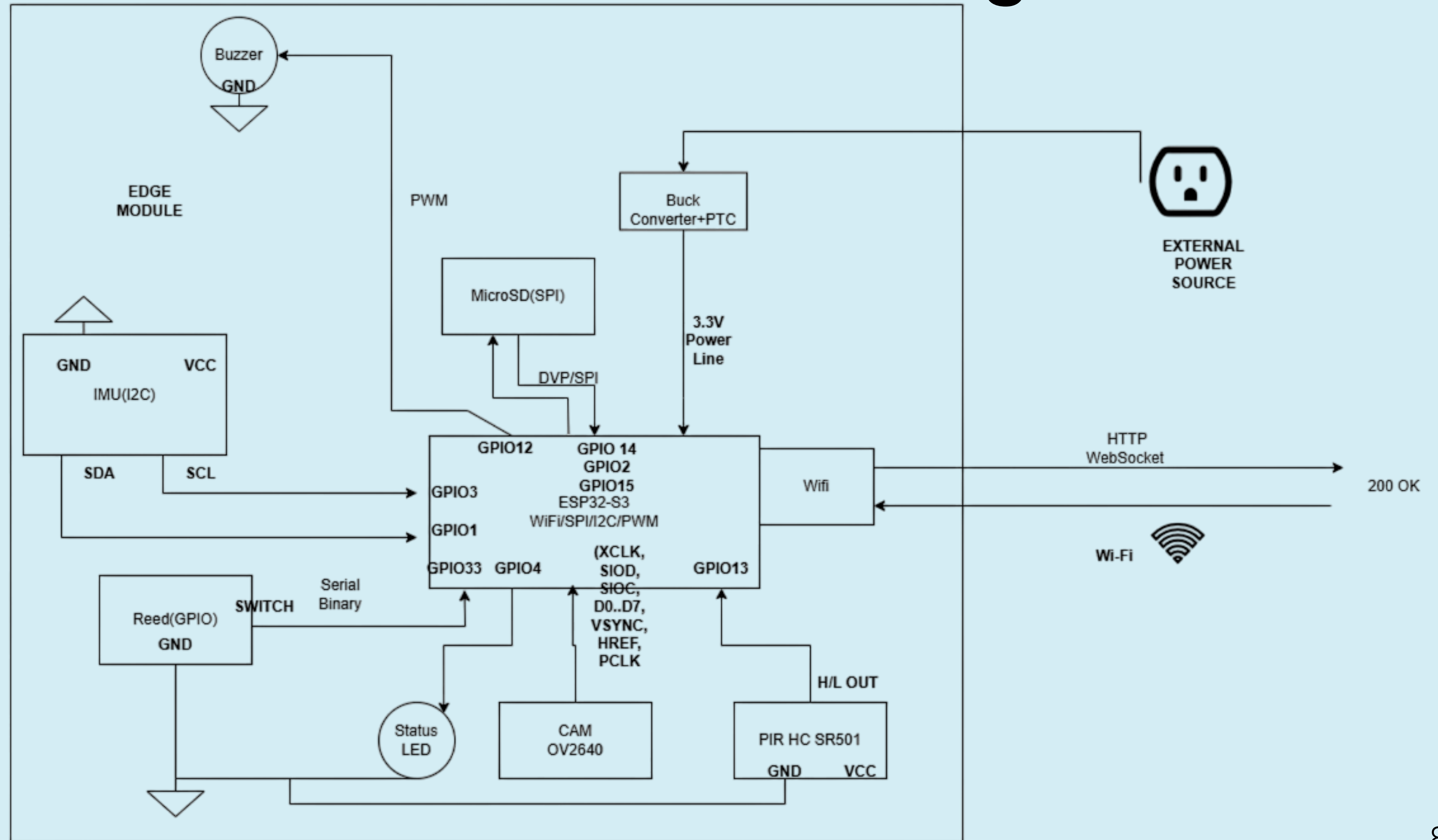


Figure 10: Edge Hardware Block Diagram

Gateway Node Hardware Block Diagram

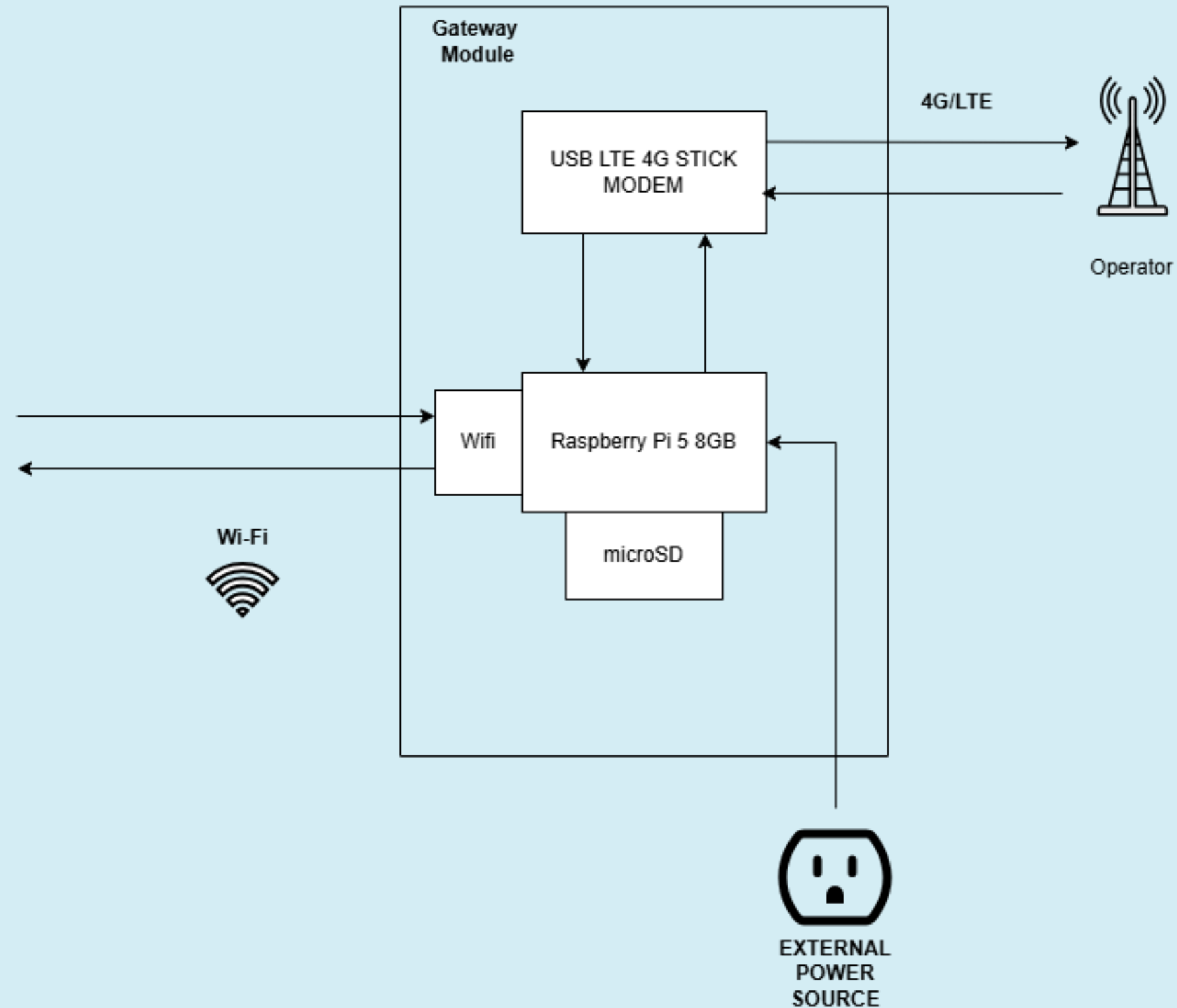


Figure 11: Total Gateway Hardware Block Diagram

Hardware Components

Edge Module:

ESP32-S3 [2]
OV2640 Camera
IMU
PIR sensor
Reed switch
Buzzer via PWM
Status LED
microSD storage

Gateway Module:

Raspberry Pi 5 [3]
USB LTE 4G modem
microSD

Edge Software

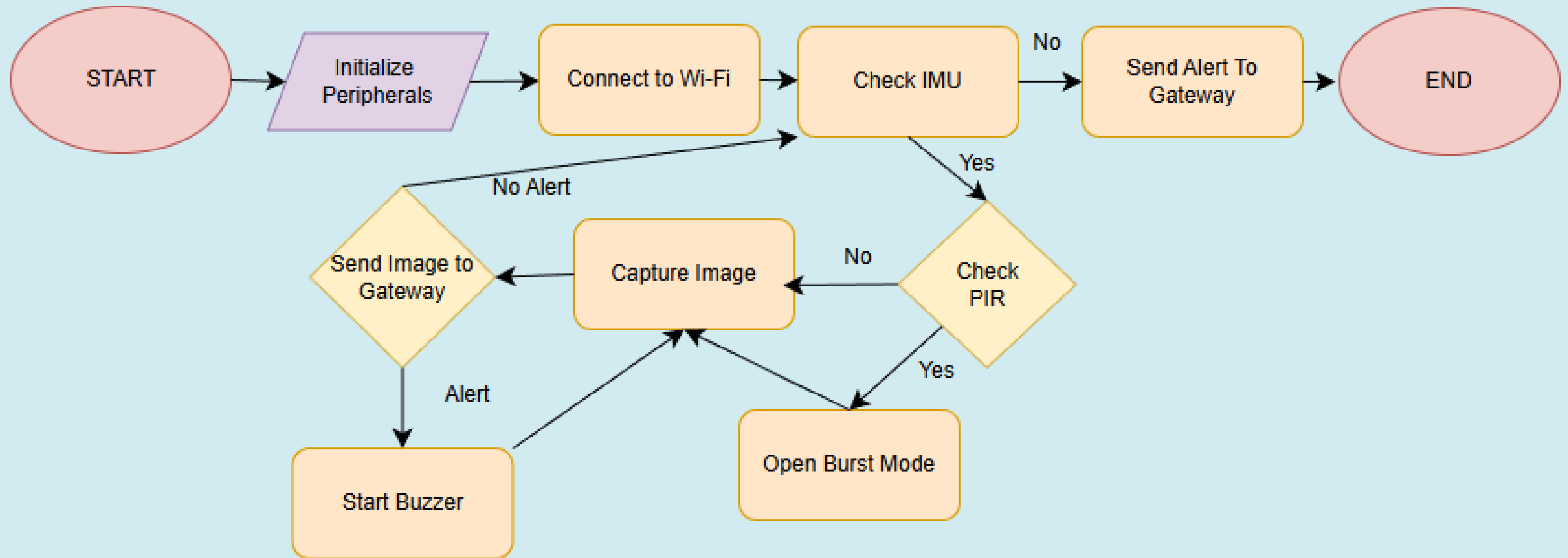


Figure 12: Edge Software Flowchart [1]

Gateway Software

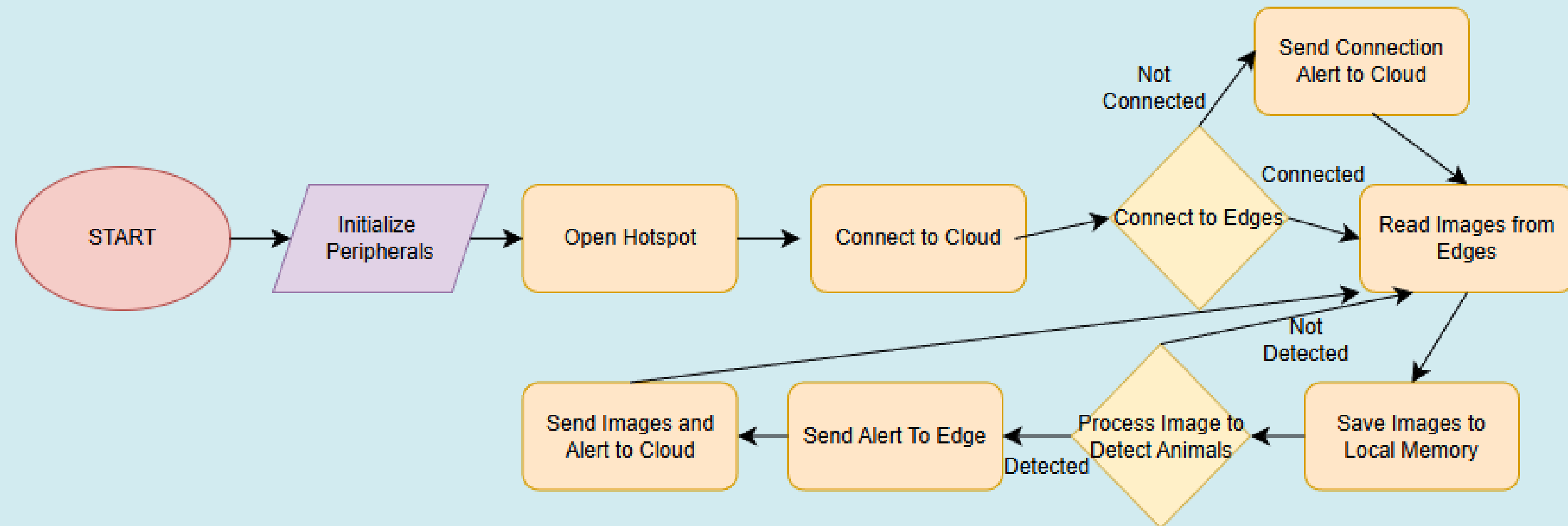


Figure 13: Gateway Software Flowchart [1]

Mobile App

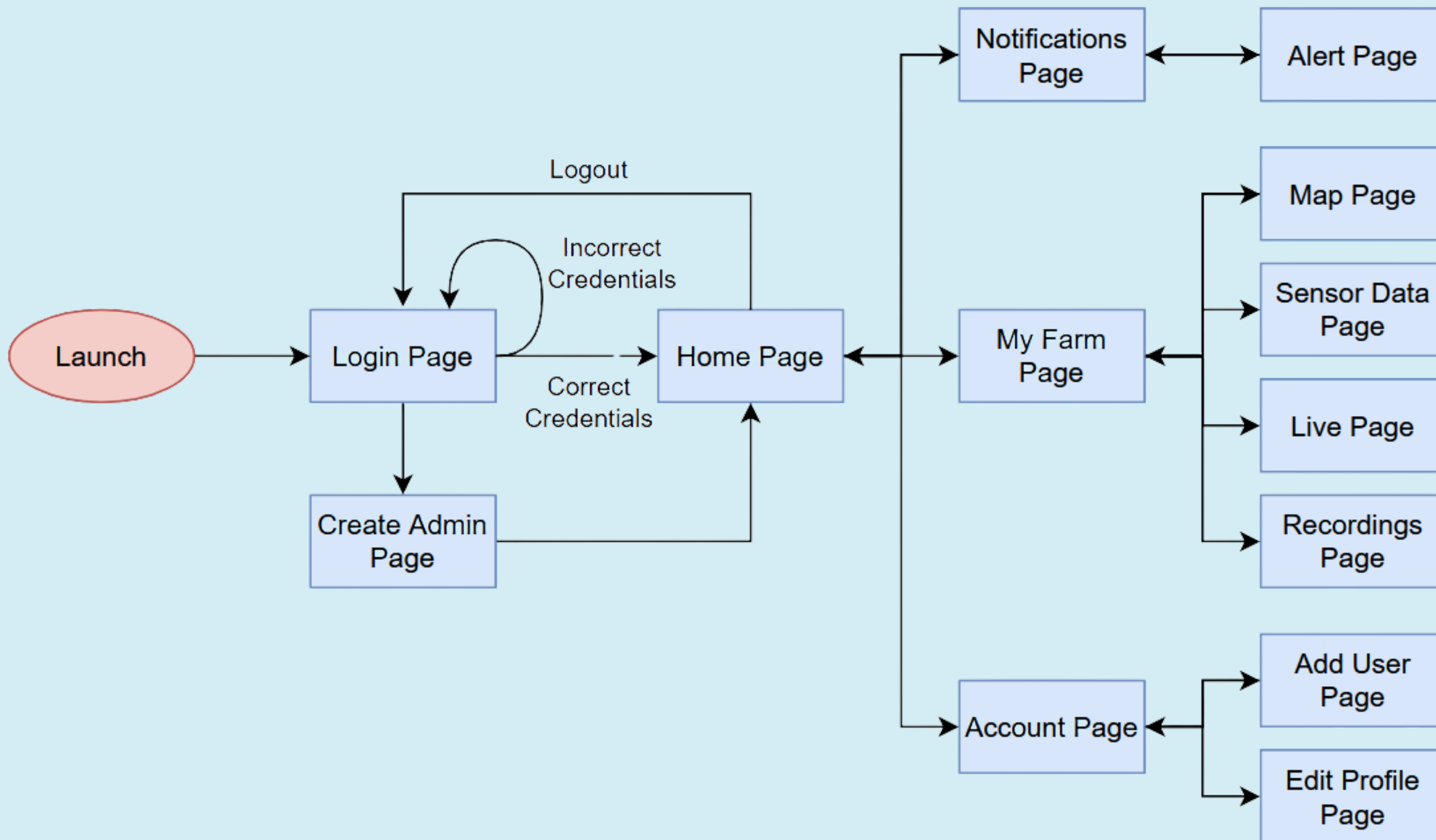


Figure 14: Mobile App Flowchart [1]

Programs and Languages Used

- Raspberry Pi OS 64Bit operating system
- Python programming language used in gateway with libraries PyTorch, Flask, and Ultralytics
- Operations in edges are done using C++ with libraries HTTPclient and esp_camera
- Mobile app is developed using Flutter UI in Dart and Android Studio

Division of Responsibilities

- Muhammed Gölcük – Edge Hardware
ESP32-CAM module, sensors, buzzer, and ESP32↔RPi communication.
- Yusuf Semih Gürkök – AI Detection
Creation of dataset, YOLOv11 training, RPi inference pipeline.
- İsmail Özgenç – Cloud & APIs
Cloud communication, data sync, REST API backend.
- Zeynep İrem İpek – Mobile App
Flutter UI, notifications, event display, remote control functions.

References

- [1] M. Gölcük, Y. S. Gürkök, İ. Özgenç, and Z. İ. İpek, FarmGate Sentinel High Level Design Document, EEE495 Electrical and Electronics Engineering Design II, Bilkent University, 2025.
- [2] ESP32 Series Datasheet, Version 4.6, Espressif Systems, 2023. Available at: <https://www.espressif.com/>
- [3] Raspberry Pi Foundation, Raspberry Pi 5 Product Brief, Raspberry Pi Ltd., 2023. Available at: <https://www.raspberrypi.com/>